

# Aidan Gardner-O'Kearny

📍 Knoxville, Tennessee    ✉ aidan.gardnerokearny@cern.ch    ☎ 503-333-6975    🔗 aidangardnokearny.github.io  
in aidangardnerokearny    🐙 agardner    🐙 aidangardnerokearny

## Education

|                                                                                                                                                                                               |                                                      |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------------|
| PhD                                                                                                                                                                                           | Univeristy Of Tennessee, Knoxville, Physics          | Aug 2025 - Present   |
| MS                                                                                                                                                                                            | Univeristy Of Tennessee, Knoxville, Physics          | Aug 2025 - Present   |
| BS                                                                                                                                                                                            | University of Oregon, Physics - Clark Honors College | Sept 2021 – Jun 2025 |
| <ul style="list-style-type: none"><li>• <b>Honors Thesis:</b> Combinatorial fake discriminatory selections for combined dE/dx and disappearing track search with the ATLAS detector</li></ul> |                                                      |                      |

## Research Experience

|                                                                                                                                                                                                                                                                          |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Holmes Group</b> - Graduate Research Assistant                                                                                                                                                                                                                        | Knoxville, TN       |
| <ul style="list-style-type: none"><li>• Worked as a part of the MAIA detector group</li><li>• Characterized photon reconstruction within PandoraPFA framework and optimized tagging criteria</li></ul>                                                                   | May 2026 - Present  |
| <b>Jeanty Group</b> - Undergraduate Research Assistant                                                                                                                                                                                                                   | Eugene, OR          |
| <ul style="list-style-type: none"><li>• Worked with Dr. Laura Jeanty as a part of the ATLAS Collaboration</li><li>• Part of upcoming dE/dx and disappearing track signature analysis for long-lived, massive, charged particles</li></ul>                                | Jun 2023 – Jun 2025 |
| <b>Brookhaven National Lab</b> - SULI Intern                                                                                                                                                                                                                             | Upton, NY           |
| <ul style="list-style-type: none"><li>• Worked with Dr. Abraham Tishelman-Charny as a part of the ATLAS Collaboration</li><li>• Member of upcoming <math>SH \rightarrow b\bar{b}\gamma\gamma</math> analysis searching for beyond Standard Model scalar bosons</li></ul> | Jun 2024 – Aug 2024 |

## Teaching Experience

|                                                                                                                                                |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>UTK Physics Department</b> - Graduate Teaching Assistant                                                                                    | Knoxville, TN        |
| <ul style="list-style-type: none"><li>• Instructor for general physics for engineers</li></ul>                                                 | Sept 2025 - Present  |
| <b>UO Tutoring and Academic Engagement Center</b> - Class Encore Mentor                                                                        | Eugene, OR           |
| <ul style="list-style-type: none"><li>• Leadership role within the Class Encore program</li></ul>                                              | Jun 2024 - Jun 2025  |
| <b>UO Tutoring and Academic Engagement Center</b> - Class Encore Leader                                                                        | Eugene, OR           |
| <ul style="list-style-type: none"><li>• Support students in introductory courses in order to increase confidence in lecture material</li></ul> | Sept 2022 - Jun 2025 |
| <b>Oregon Pathways to Industry Research Careers</b> - OPIRC Tutor                                                                              | Eugene, OR           |
| <ul style="list-style-type: none"><li>• Multivariable calculus tutor</li></ul>                                                                 | Apr 2025 - Jun 2025  |
| <b>UO Physics Department</b> - Introductory Physics Learning Assistant                                                                         | Eugene, OR           |
| <ul style="list-style-type: none"><li>• Member of the general physics educational team</li></ul>                                               | Apr 2024 - Jun 2024  |

## Presentations

|                                                                                                                                                                                         |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Combinatorial fake discriminatory selections for combined dE/dx and disappearing track search</b>                                                                                    | May 2025 |
| <ul style="list-style-type: none"><li>• UO Undergraduate Research Symposium</li><li>• Oral presentation</li><li>• Poster presentation</li><li>• <b>Aidan Gardner-O'Kearny</b></li></ul> |          |

|                                                                                                                                                                                                             |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>dE/dx and disappearing track background discrimination studies</b>                                                                                                                                       | April 2025  |
| <ul style="list-style-type: none"> <li>• UO Clark Honors College Three Minute Thesis</li> <li>• Oral presentation</li> <li>• <b>Aidan Gardner-O'Kearny</b></li> </ul>                                       |             |
| <b>Background ionization behavior for combined dE/dx and disappearing track analysis in the ATLAS detector</b>                                                                                              | March 2025  |
| <ul style="list-style-type: none"> <li>• APS Global Summit</li> <li>• Oral Presentation</li> <li>• <b>Aidan Gardner-O'Kearny</b>, Laura Jeanty, Nathan Young</li> </ul>                                     |             |
| <b>Comparing resolved and boosted jet identification algorithms to search for beyond the Standard Model scalar bosons with the ATLAS detector</b>                                                           | August 2024 |
| <ul style="list-style-type: none"> <li>• Brookhaven SULI Intern Research Fair</li> <li>• Poster Presentation</li> <li>• <b>Aidan Gardner-O'Kearny</b>, Abraham Tishelman-Charny, Elizabeth Brost</li> </ul> |             |
| <b>Background ionization behavior for combined dE/dx and disappearing track analysis</b>                                                                                                                    | May 2024    |
| <ul style="list-style-type: none"> <li>• UO Undergraduate Research Symposium</li> <li>• Poster Presentation</li> <li>• <b>Aidan Gardner-O'Kearny</b>, Laura Jeanty, Nathan Young</li> </ul>                 |             |

## Awards

---

|                                                                                                                                                   |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>UO Physics Department Weiser Research Award</b>                                                                                                | June 2025 |
| <b>UO Physics Department Poster Award - 2nd Place</b>                                                                                             | June 2025 |
| <ul style="list-style-type: none"> <li>• Combinatorial fake discriminatory selections for combined dE/dx and disappearing track search</li> </ul> |           |
| <b>UTK Graduate Fellowship</b>                                                                                                                    | Feb 2025  |
| <b>UO CURE Travel Grant Awardee</b>                                                                                                               | Nov 2024  |
| <ul style="list-style-type: none"> <li>• Award given to support undergraduates presenting their research at a relevant conference</li> </ul>      |           |
| <b>UO Summit Scholar</b>                                                                                                                          | Sept 2021 |
| <ul style="list-style-type: none"> <li>• Academic excellence scholarship</li> </ul>                                                               |           |

## Outreach

---

|                                                                                                                                                                                        |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Eugene Science Center</b>                                                                                                                                                           | Nov 2024 |
| <ul style="list-style-type: none"> <li>• Organized an outreach event for elementary school students</li> <li>• Provided undergraduate students opportunity to perform demos</li> </ul> |          |

## Membership

---

|                                                                          |                     |
|--------------------------------------------------------------------------|---------------------|
| <b>Phi Beta Kappa</b>                                                    | June 2025           |
| <b>American Physical Society</b>                                         | Oct 2024 - Present  |
| <ul style="list-style-type: none"> <li>• Student Member</li> </ul>       |                     |
| <b>Society Of Physics Students</b>                                       | Jun 2023 - Jun 2025 |
| <ul style="list-style-type: none"> <li>• UO Chapter Treasurer</li> </ul> |                     |

## Skills

---

**Languages:** Python, C++, JULIA, Bash, T<sub>E</sub>X/Markdown, Mathematica

**Technologies:** Linux, Windows, ROOT, Matplotlib/MPLHEP, Git, PandoraPFA, Google And Microsoft Office Suite